

رقم الجلوس:

الفصل:

الإسم:

Mention the scientific term

- (1) It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field
- (2) It is the ability of medium to penetrate the magnetic flux lines
- (3) It is the point at which the total magnetic flux density vanishes
- (4) It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field
- (5) It is the magnetic torque acting on a coil carrying current placed parallel to a uniform magnetic field of flux density of 1 tesla
- (6) It is the total number of magnetic flux lines passing through a surface
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Mention the application of

- (7) ampere's right-hand rule
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- (8) magnetic torque
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Mention the factor affecting on

- (9-10) magnetic force (two factors)
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(11) magnetic field due to a current in circular coil (one factor)

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(12) Direction of magnetic field due to a current in straight wire (one factor)

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.....

solve

(13-14) A rectangular loop has dimensions of 0.50m and 0.60m. The values of magnetic field are 0.02T with an angle of 30°. Calculate the magnetic flux at the surface.

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(15-18) A solenoid of length 20 cm and 200 turns carries an electric current of 0.5 A. Find the magnetic flux density at the middle of its axis if: $((\mu_{\text{iron}} = 2 \times 10^{-3} \text{ Wb/A}\cdot\text{m})-(\mu_{\text{air}} = 4\pi \times 10^{-7} \text{ Wb/A}\cdot\text{m}))$

(a) the medium is air.

(b) an iron rod is inserted inside the coil.

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Put (T) for true and (F) for false

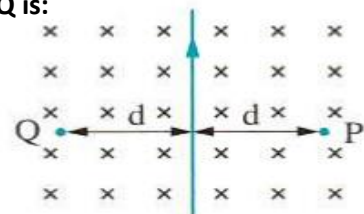
(19) In the opposite figure, a straight wire is placed perpendicular to a uniform magnetic field of flux density B that is directed into the page. When an electric current of intensity I passes in the wire, the total magnetic flux density at point P becomes 3B, so the net magnetic flux density at point Q is:

(a) zero

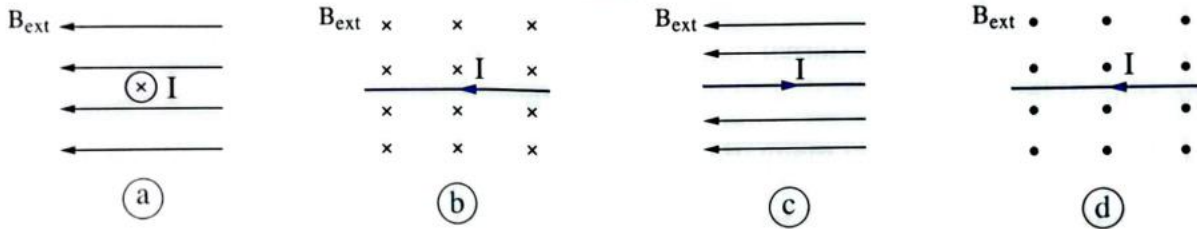
(b) B

(c) 2B

(d) 3B

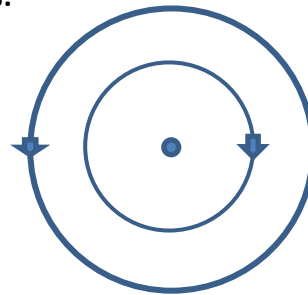


(20) In which of the following cases the wire is not affected by a magnetic force?

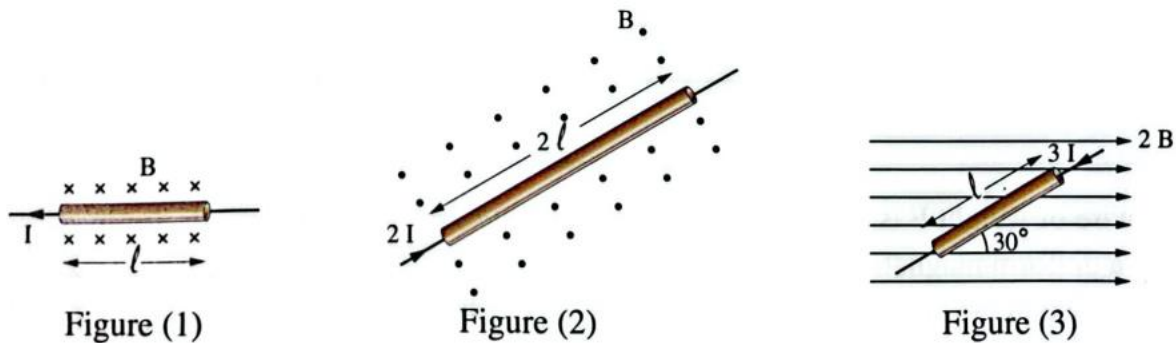


(21) Two concentric metal rings in one plane, each carries a current of intensity (I) as shown in figure, the direction of the magnetic flux at the common center (m) is:

- (a) Rightwards
- (b) leftwards
- (c) into the page
- (d) out of the page



(22) In each of the following figures, a straight wire carries a direct electric current while being in a uniform magnetic field:



Which of the following represents the correct order of the values of the magnetic forces which act on the wires?

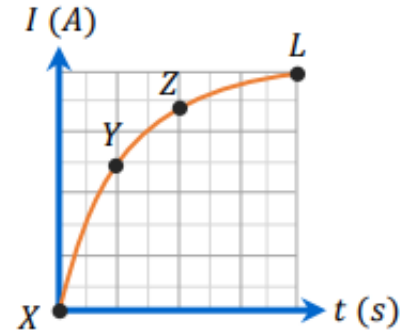
- a) $F_1 > F_2 > F_3$
- b) $F_2 > F_1 > F_3$
- c) $F_2 > F_3 > F_1$
- d) $F_3 > F_1 > F_2$

(23) Permeability measured by

- (a) $T.m/A$
- (b) $A.m^2$
- (c) $Wb/A.m^2$
- (d) $Wb/m.A^2$

(24) The opposite figure represents the growth of electric current in a solenoid at which point the magnetic flux density will be the greatest

- (a) X (b) Y
(c) Z (d) L



(25) Rule to determine the direction of the magnetic field resulting from an electric current in a circular coil

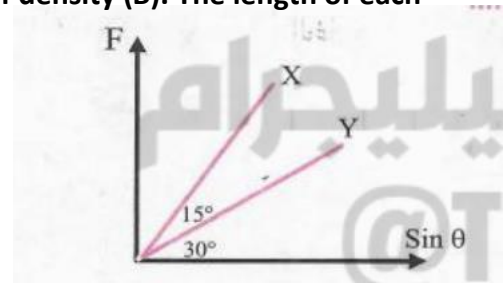
- (a) Fleming right hand rule (b) Fleming left hand rule
(c) polarity (d) Ampere left hand rule

(26) A straight wire of length (l) is made into a circular coil of radius (r). A current of intensity (i) passes through it, creating a magnetic field of density (B) at its center. If the same wire is reshaped into another circular coil of radius ($\frac{2r}{3}$) with the same current intensity, the magnetic field density at its center becomes...

- a) $\frac{3B}{2}$ b) $\frac{2B}{3}$
c) $\frac{9B}{4}$ d) $\frac{4B}{9}$

(26) Two conductors, X and Y, are placed in a magnetic field of density (B). The length of each conductor is (l). from the figure what is the ratio of $\frac{I_x}{I_y}$

- a) $\sqrt{3}$ b) $\frac{1}{\sqrt{3}}$
c) $\sqrt{2}$ d) $\frac{1}{\sqrt{1}}$



(27) A person is standing at a distance (d) from an electric power cable; hence he gets affected by a magnetic flux of density B . If this person moves to a place at a distance ($\frac{2d}{3}$) from the cable, the magnetic flux density experienced by him will increase by...

- a) 25% b) 33.3%
c) 50% d) 66.7%

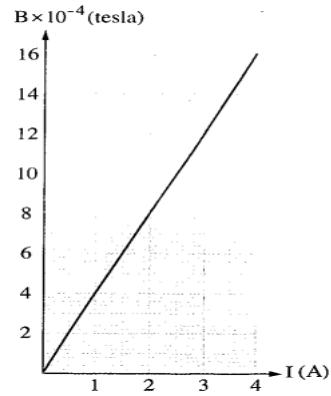
(28) Magnetic lines come from..... looks like long bar magnet

- a) Solenoid coil
- b) Circular coil
- c) Straight wire
- d) All the previous

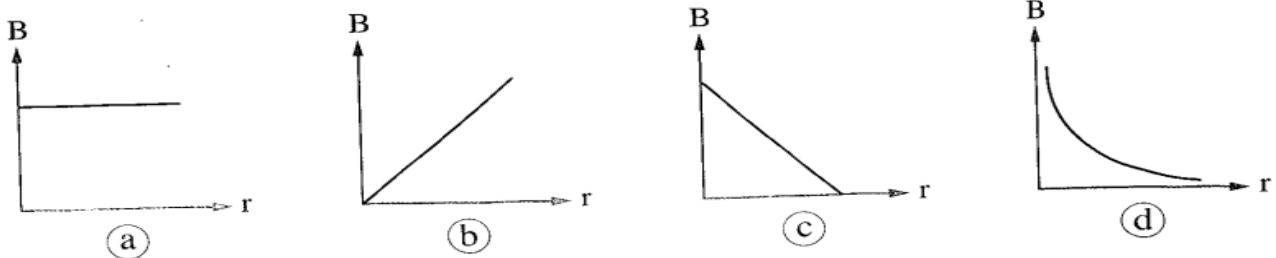
(29) The graph represents the relation between the magnetic flux density (B) inside a solenoid and the current intensity (I) that flows in the solenoid.

The number of turns per meter for this solenoid equals
(where: $\mu_{\text{air}} = 4\pi \times 10^{-7} \text{ Wb/A}\cdot\text{m}$)

- (a) 215.2 turns/meter
- (b) 250.5 turns/meter
- (c) 318.18 turns/meter
- (d) 341.4 turns/meter



(30) The graph represents the relation between radius (r) of a circular coil and the magnitude of the magnetic flux density (B) at the center of the coil



Correct the underlined in the following statement

(31) The maximum value of magnetic flux cutting a certain area when the magnetic flux lines is parallel to that area.
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(32) The magnetic field density generated from a solenoid carrying-current is inversely proportional to the distance to that point.
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(33) The rule used to determine the direction of magnetic flux density at a point nearer to a wire have an electric current pass through it is polarity.
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(34) The magnetic flux density at center of a circular coil carrying a current decrease by decreasing the cross-sectional area of the coil.
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